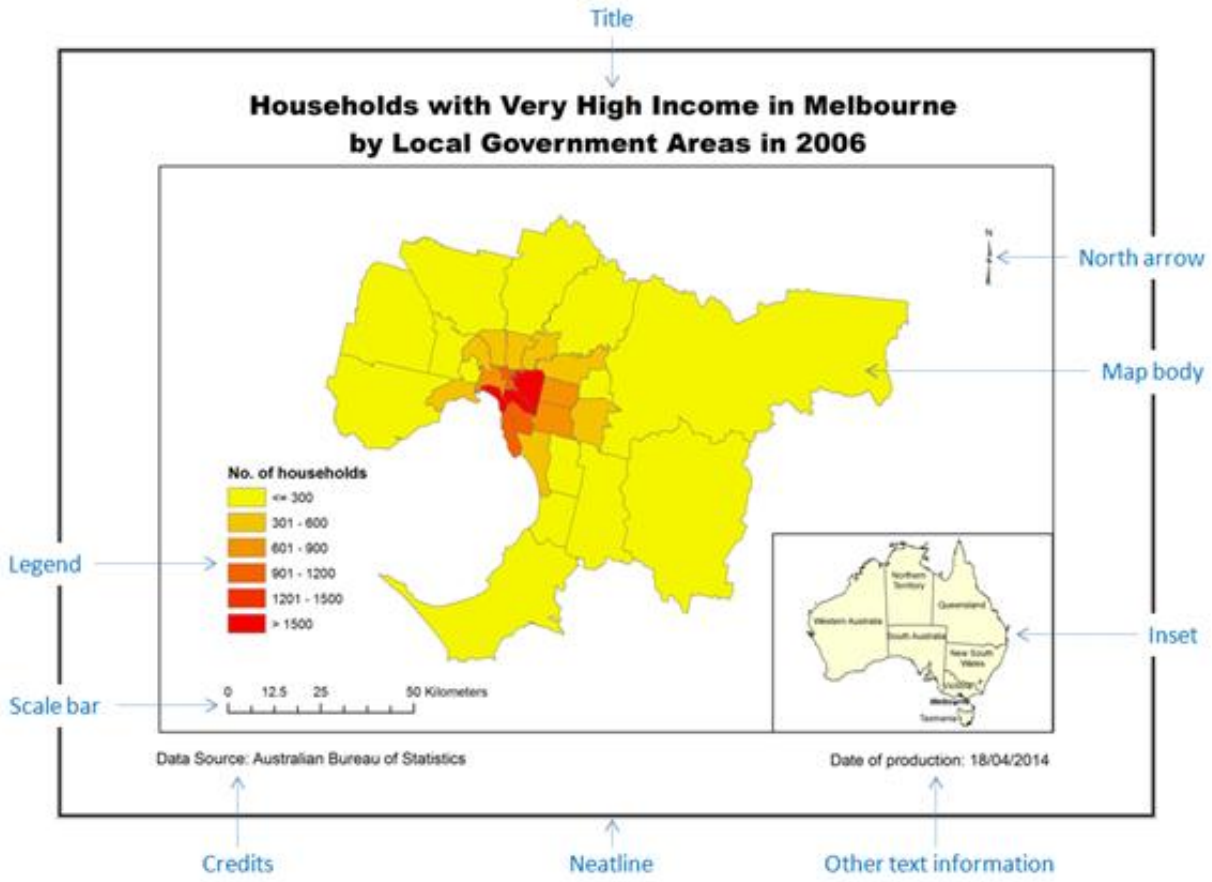
Spatial visualization

Lecture content

- The basic elements of a map.
- Map types and projections.
- How to design a map and effective layout.
- Visual hierarchy.
- Map lettering and placement.
- Multivariate maps.
- Animated maps.

Cartographic elements



Map types

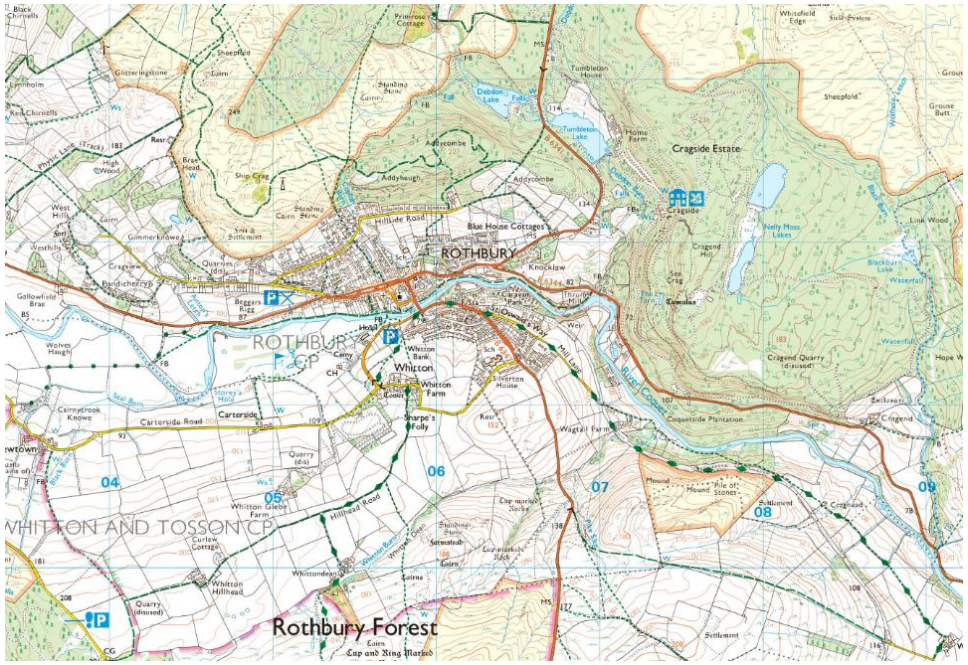
General purpose maps

- provide geographical information for reference purposes
  - topographic maps - comprehensive reference information about a region
  - base map - a foundation for superimposing additional layers of information

Thematic maps

- map the spatial distribution of a single attribute or variable or the relationship among several variables

General purpose maps



Topographic map *– reference information about an area*



Base map *– context, orientation and background for data layers*

Map types

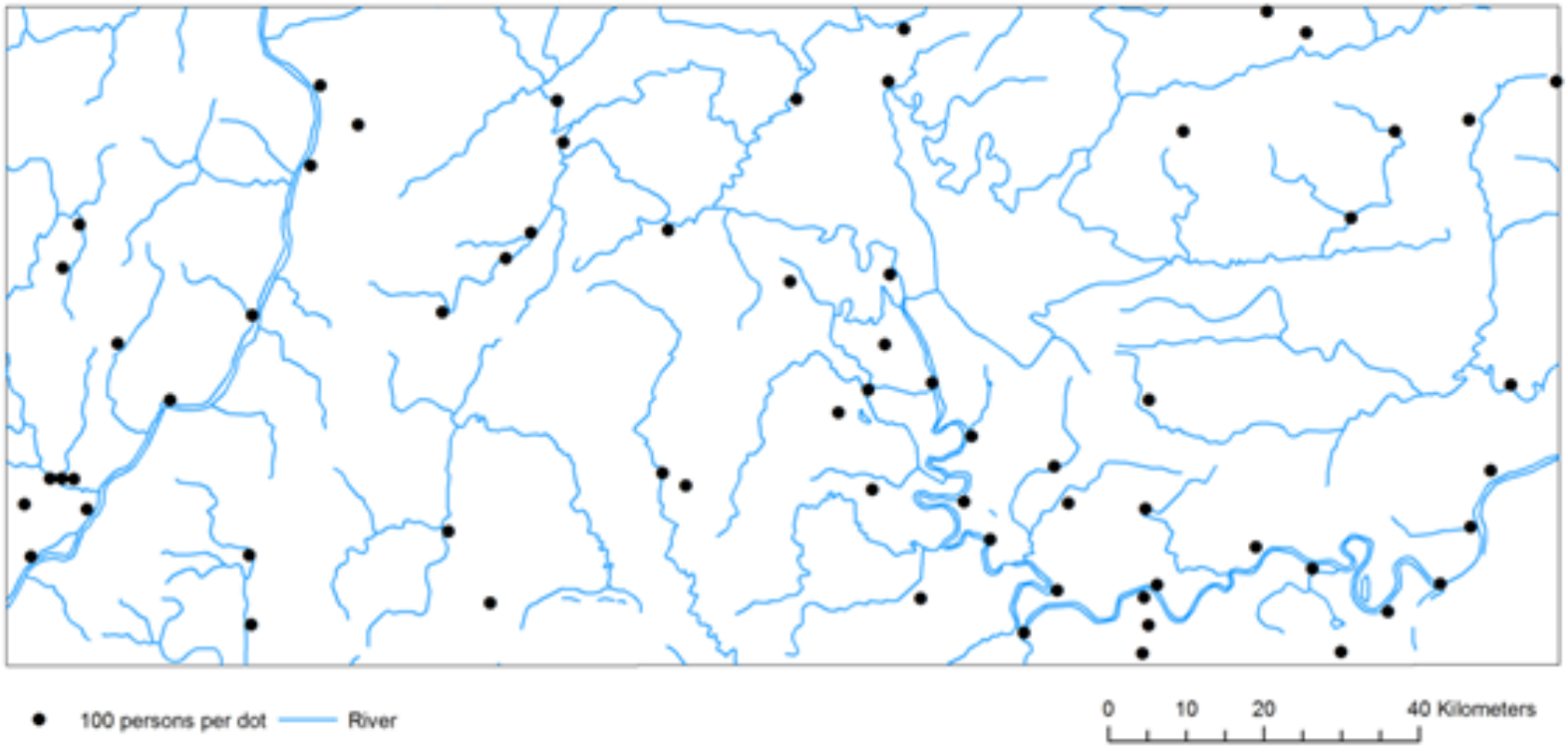
General purpose maps

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Thematic maps

- map the spatial distribution of a single attribute or variable or the relationship among several variables

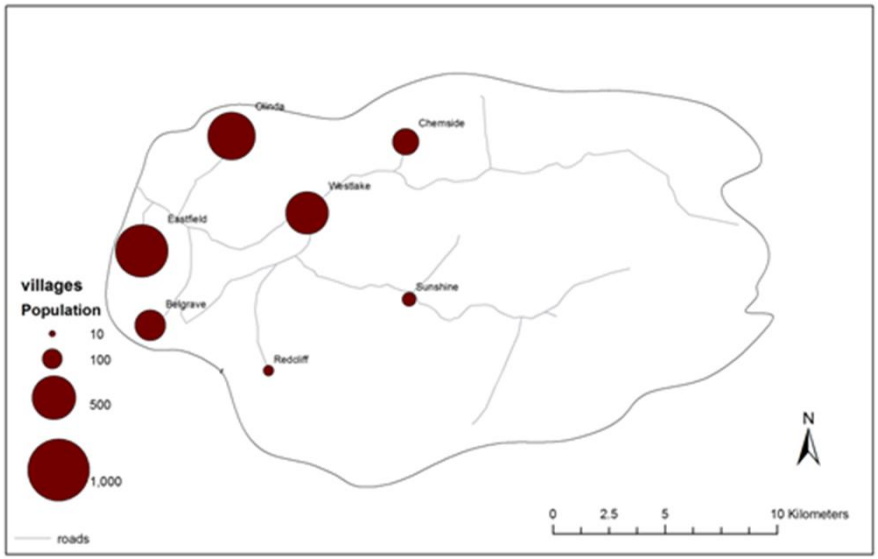
Dot map



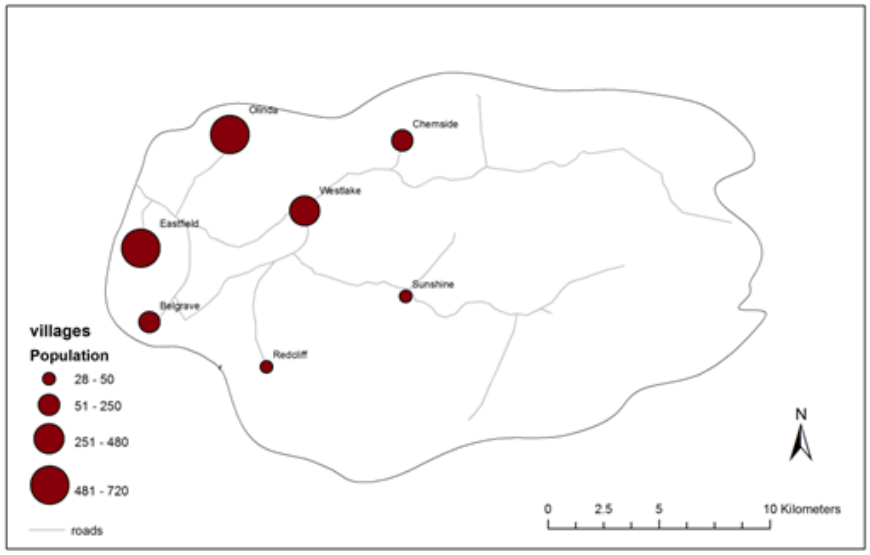
Represent the spatial distribution of point features – dots are values (not a surface maps)

Proportional and graduated symbol maps

Proportional

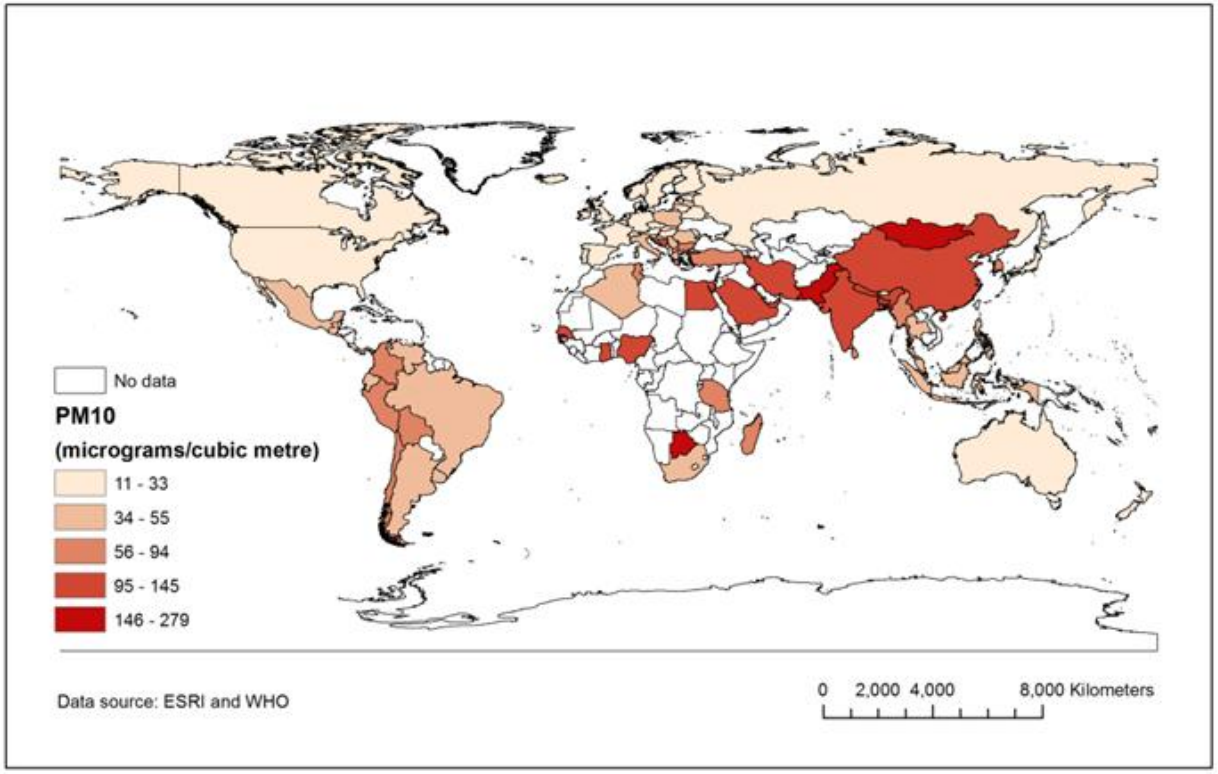


Graduated



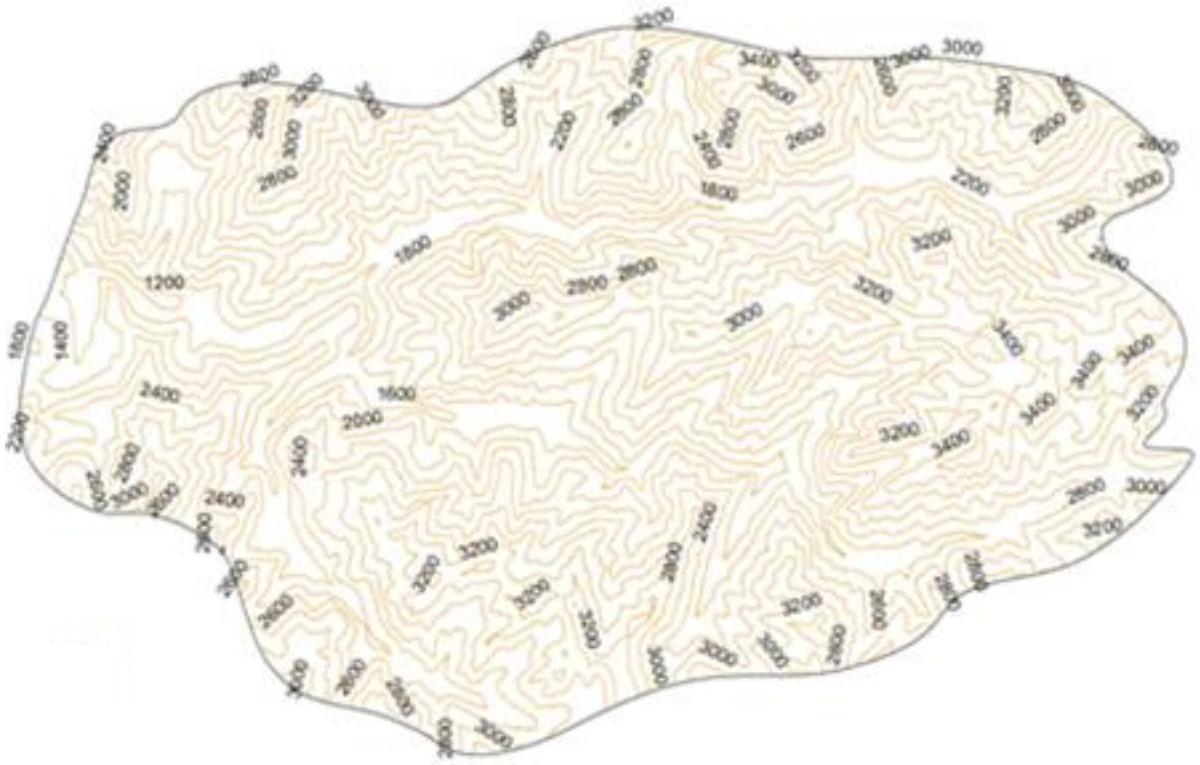
Scaled symbols to represent quantitative values of geographical features, using ratios (values or classes).

Choropleth map



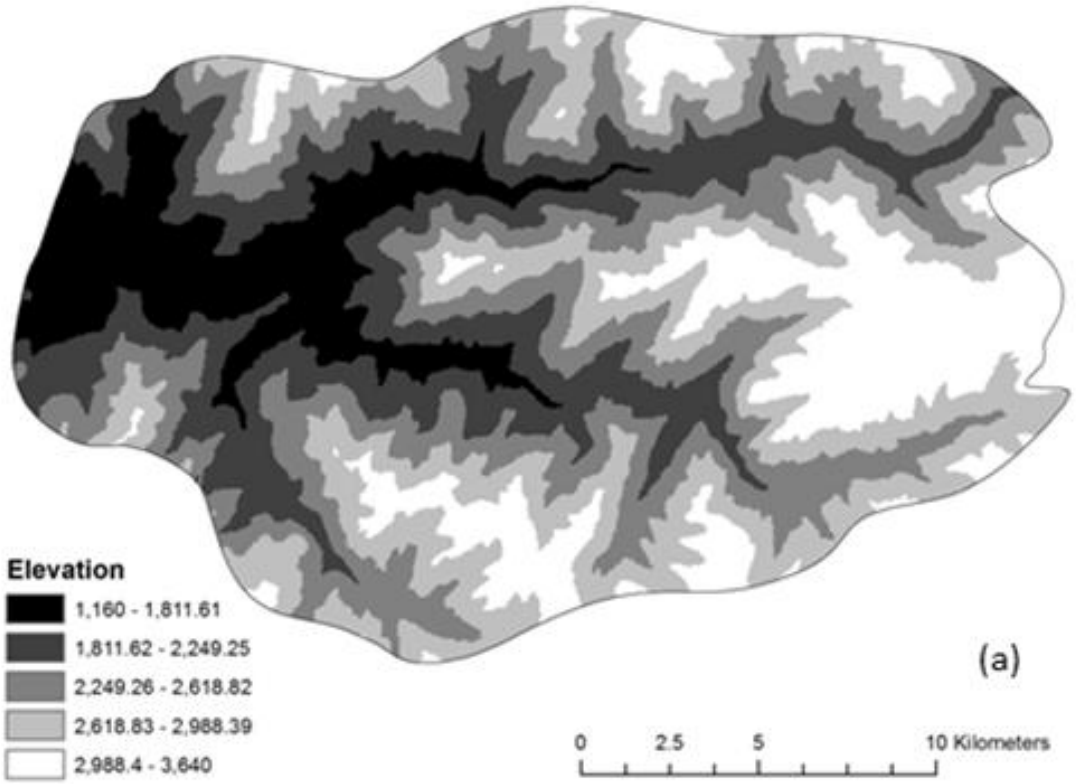
Data are classified into ranges or classes, then shading or colouring each unit area using graduated colours.

Isoline map



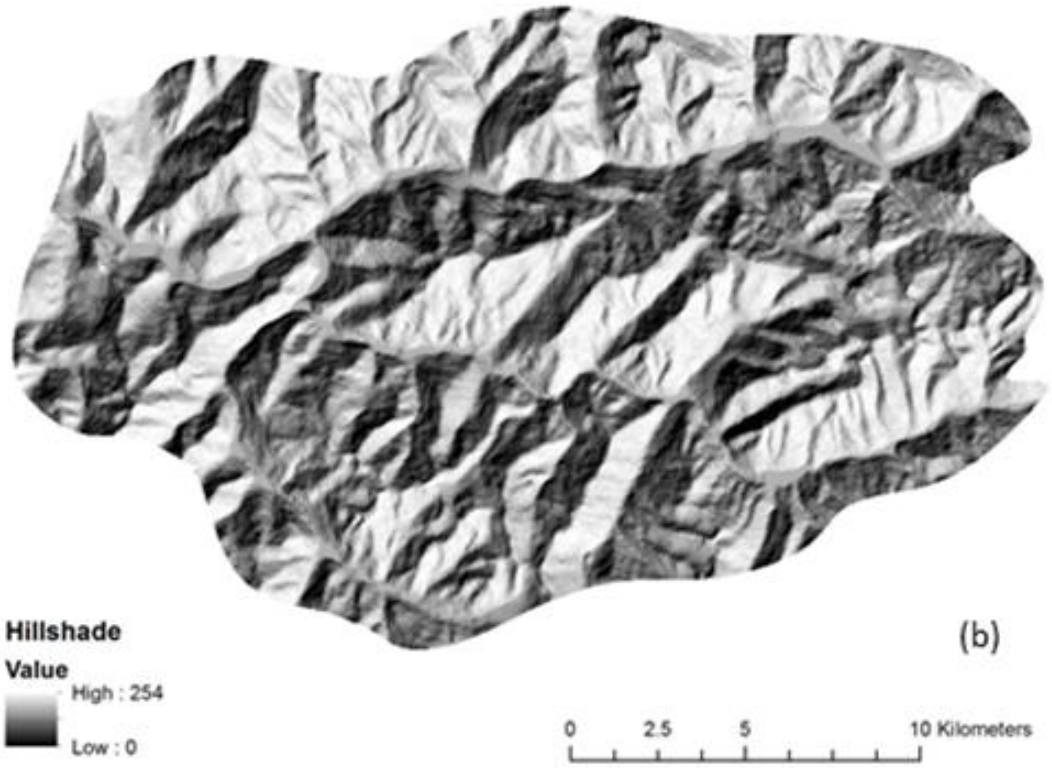
Isoline maps consist of lines joining points of equal value of a continuous variable

Hypsometric map



Shading the space between contour lines, which represents a terrain surface as a series of elevation zones.

Hill-shaded map



2D representation of 3D terrain surfaces produced by simulating the appearance of the effects of sunlight illuminating the surface of the land.

Map projections by preserved property (area, shape, direction, distance)

Equal area projection (area) preserve areas and are used for mapping distributions such as population density, land cover, geology or climatic regions.

Conformal map projections (shape and direction) e.g. geological structural data, isobars and isotherms, wind directions, ocean currents.

Equidistant projections (distances) e.g. wildfire spreading from ignition point, volcanic ash dispersal from a volcano, pollution dispersal from a point source.

Projections are a compromise between distortion, accuracy, and aesthetics.

Map design

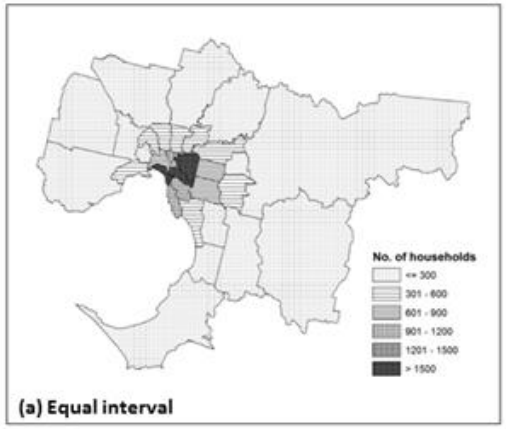
Selection: selecting the geographical area, geographical features, map scale, projection, data variables, etc.

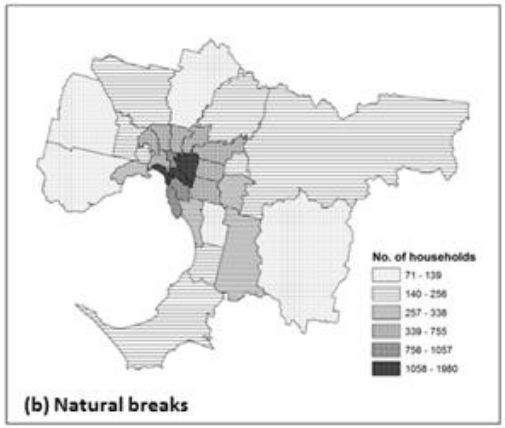
Classification: grouping and categorising data

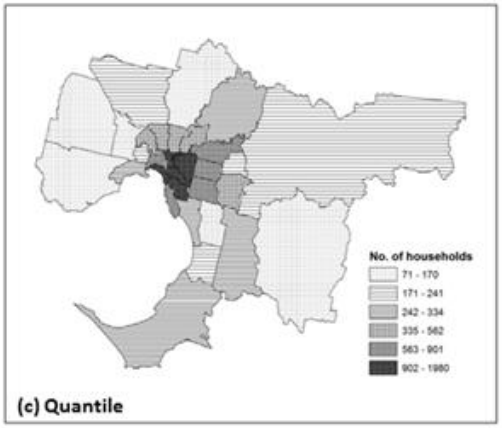
Symbolisation: selecting symbols and visual variables to represent data in a logical way

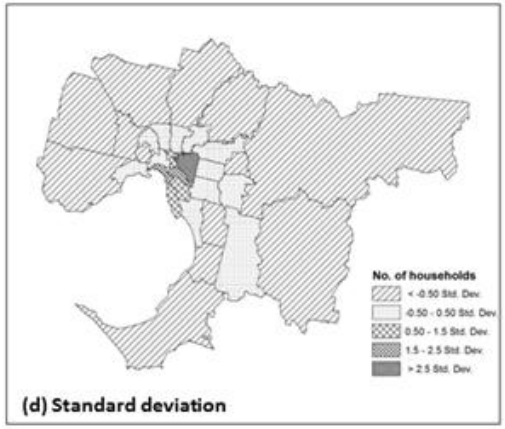
Layout design: organising disparate cartographic elements coherently on a page

Data classification



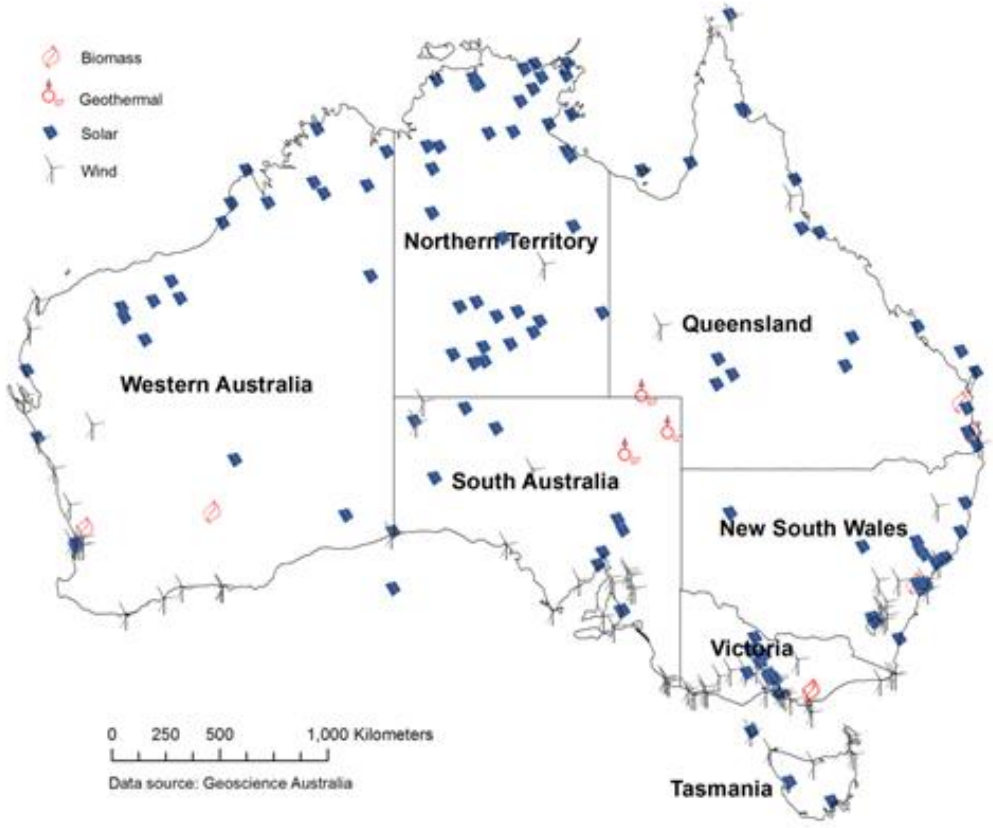






Classification involves grouping similar features into classes (5-8 max) to reduce complexity and organise data. (a) equal size ranges; (b) identifies natural patterns; (c) equal number of features in each class, irregular class sizes; (d) class breaks are standard deviations from the mean.

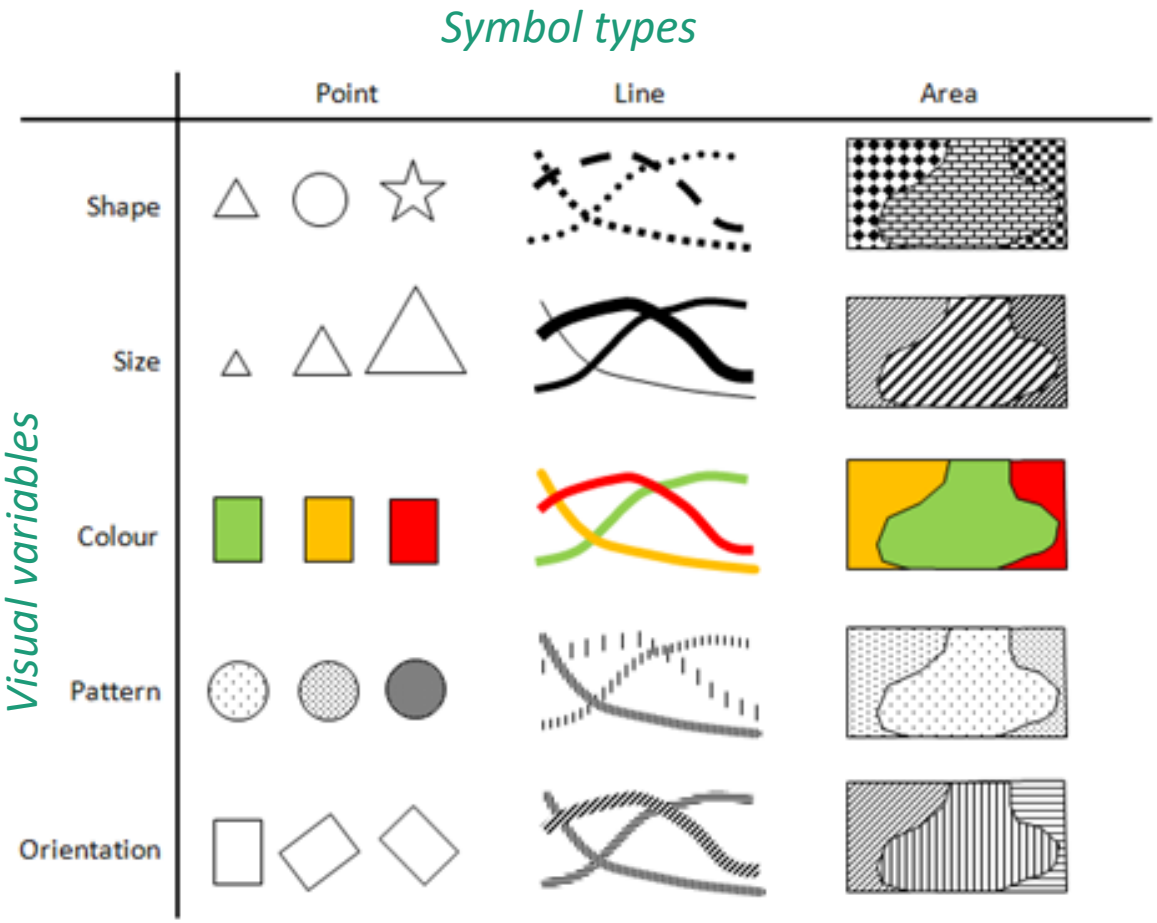
Symbolisation: use of shape



Choosing symbols and using visual variables to represent the geographical features. Simple, self-explanatory symbols are most appropriate.

Symbol geometry is qualitative not quantitative.

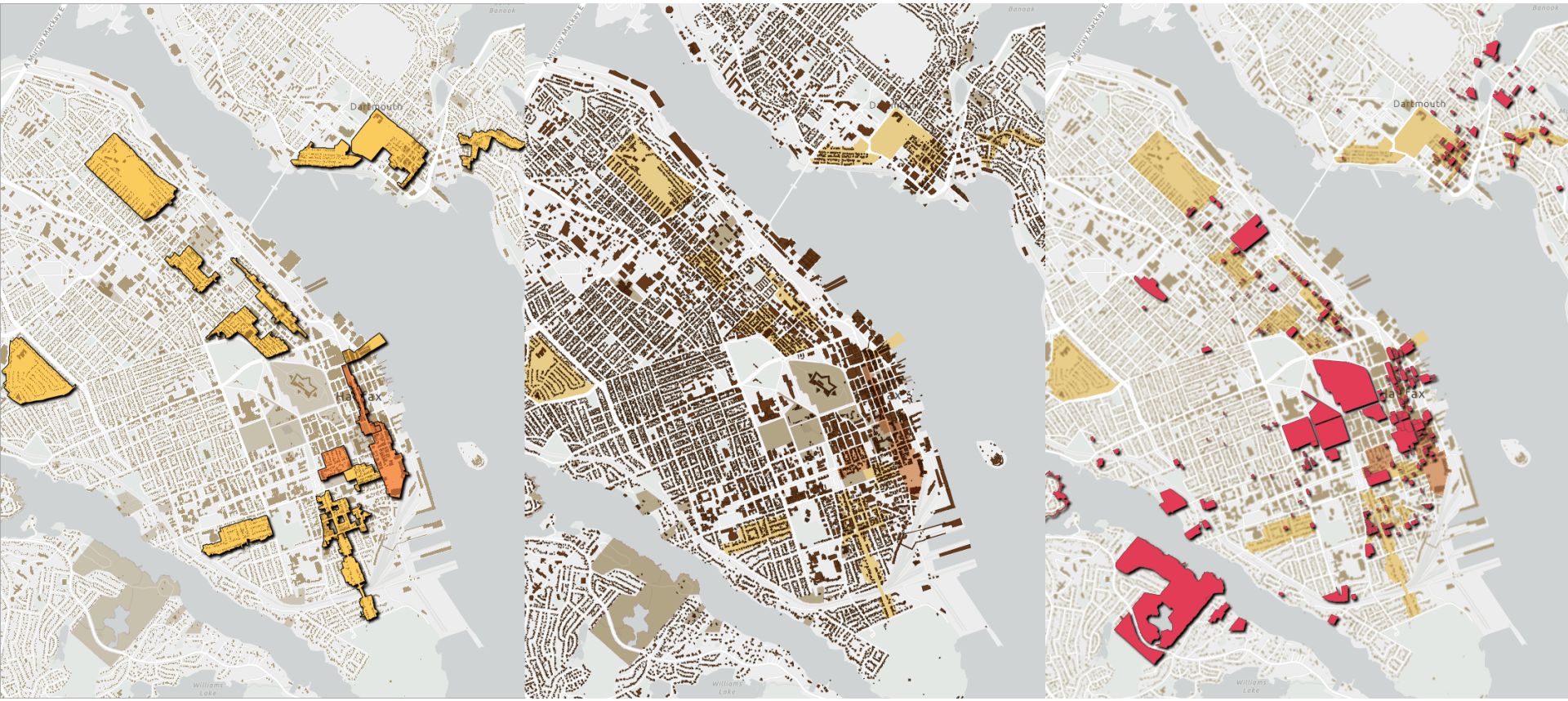
Symbolisation: visual variables



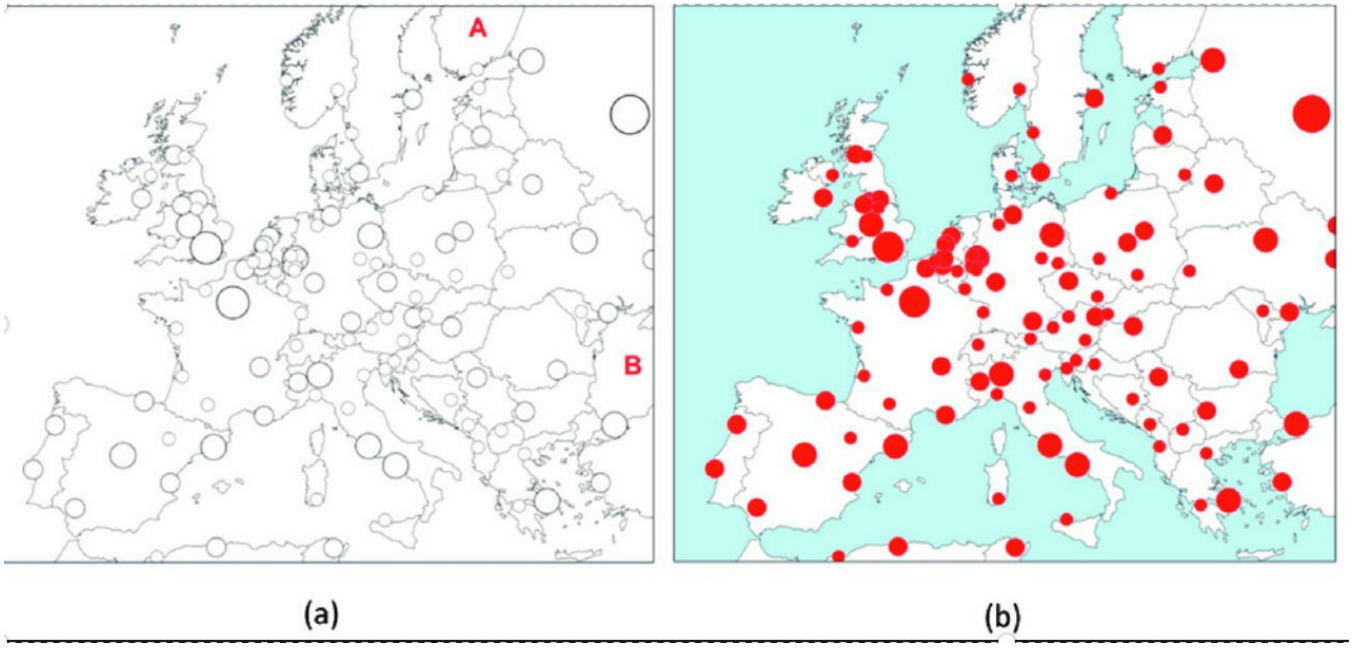
Visual hierarchy

- The perceptual organisation of mapped features.
- Uses a visual hierarchy:
  - the foreground features are most visually prominent, which are the primary features;
  - the secondary and supporting features have decreasing visual importance, placed in the lower levels.
- Established by visual contrasts or differences in colour intensity, colour hue, shading pattern, and size of map symbols.

Visual hierarchy



Stereogrammic method



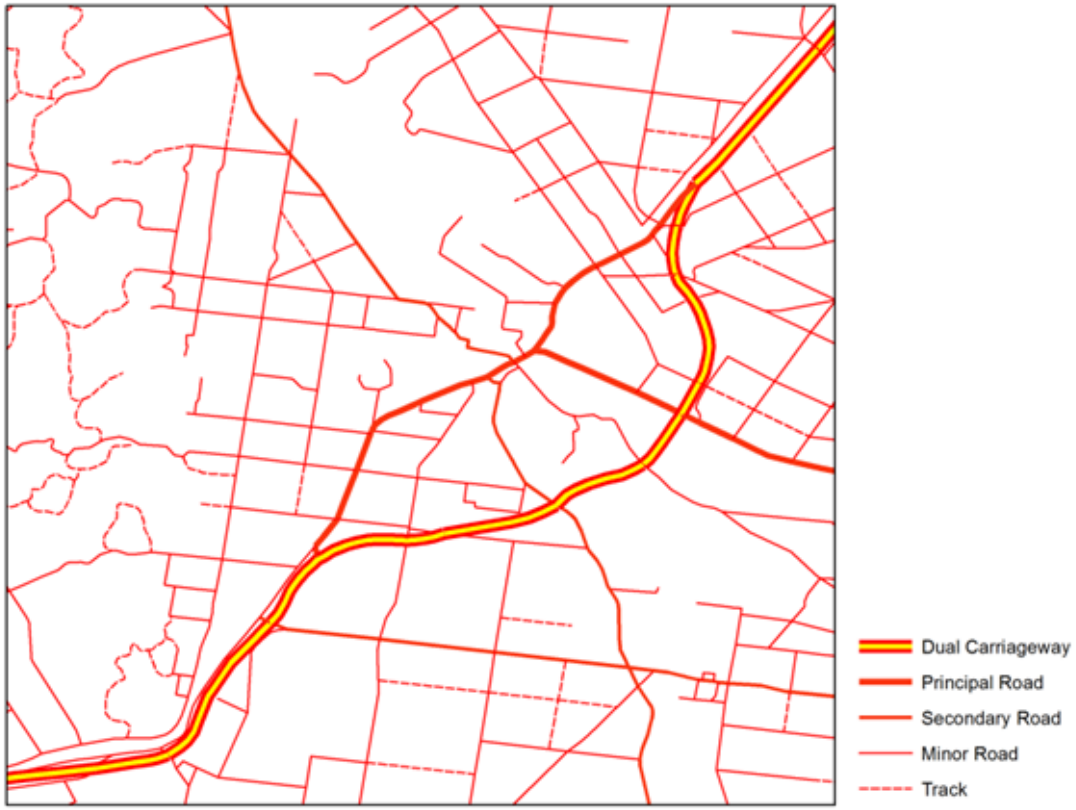
Creates a stereogrammic hierarchy that makes the important features appear above the background base data layers.

Toxic chemical pollution found in river and groundwater sites across England



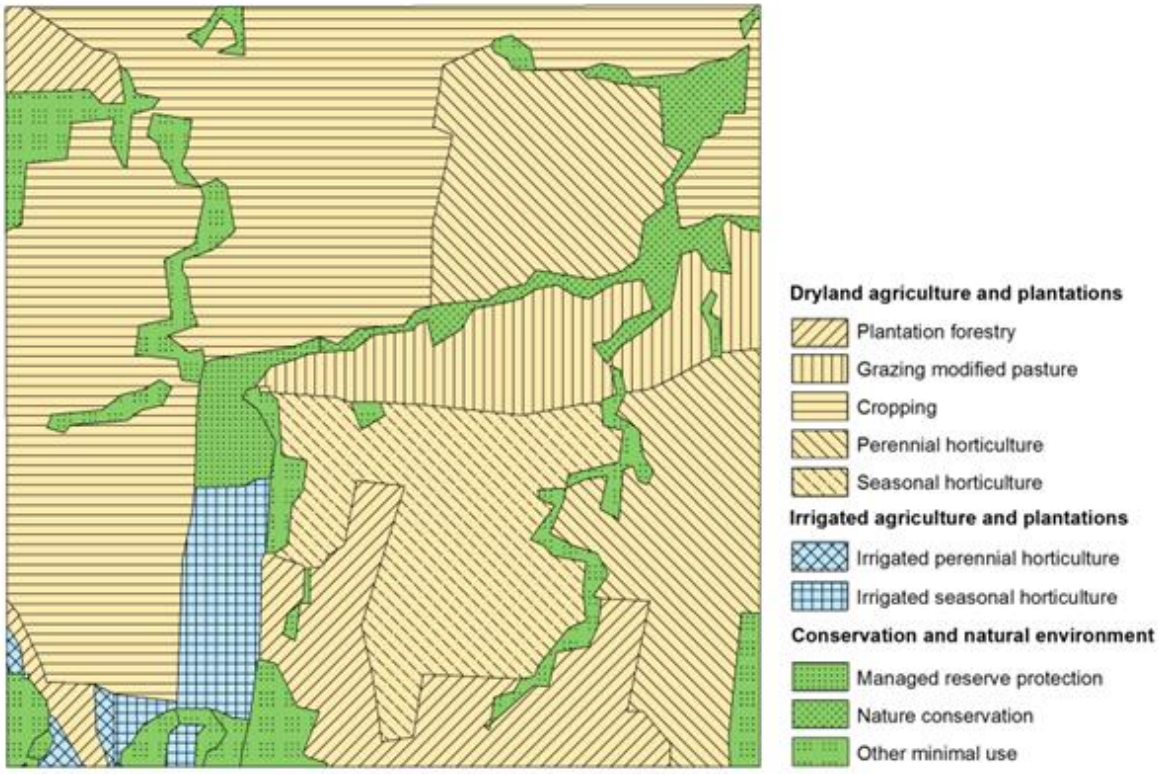


Extensional method



The extensional method aims to show a hierarchy of importance among features.

Subdivisional method

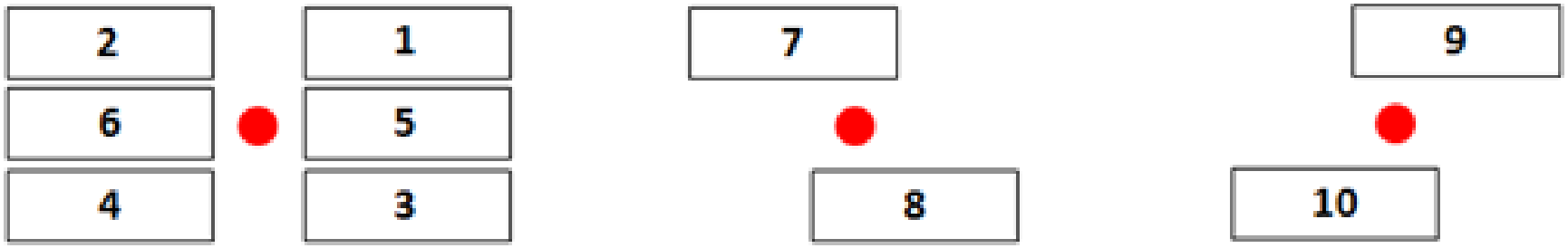


The subdivisional method uses colour and pattern to distinguish between classes and subclasses of area features.

Principles for map lettering

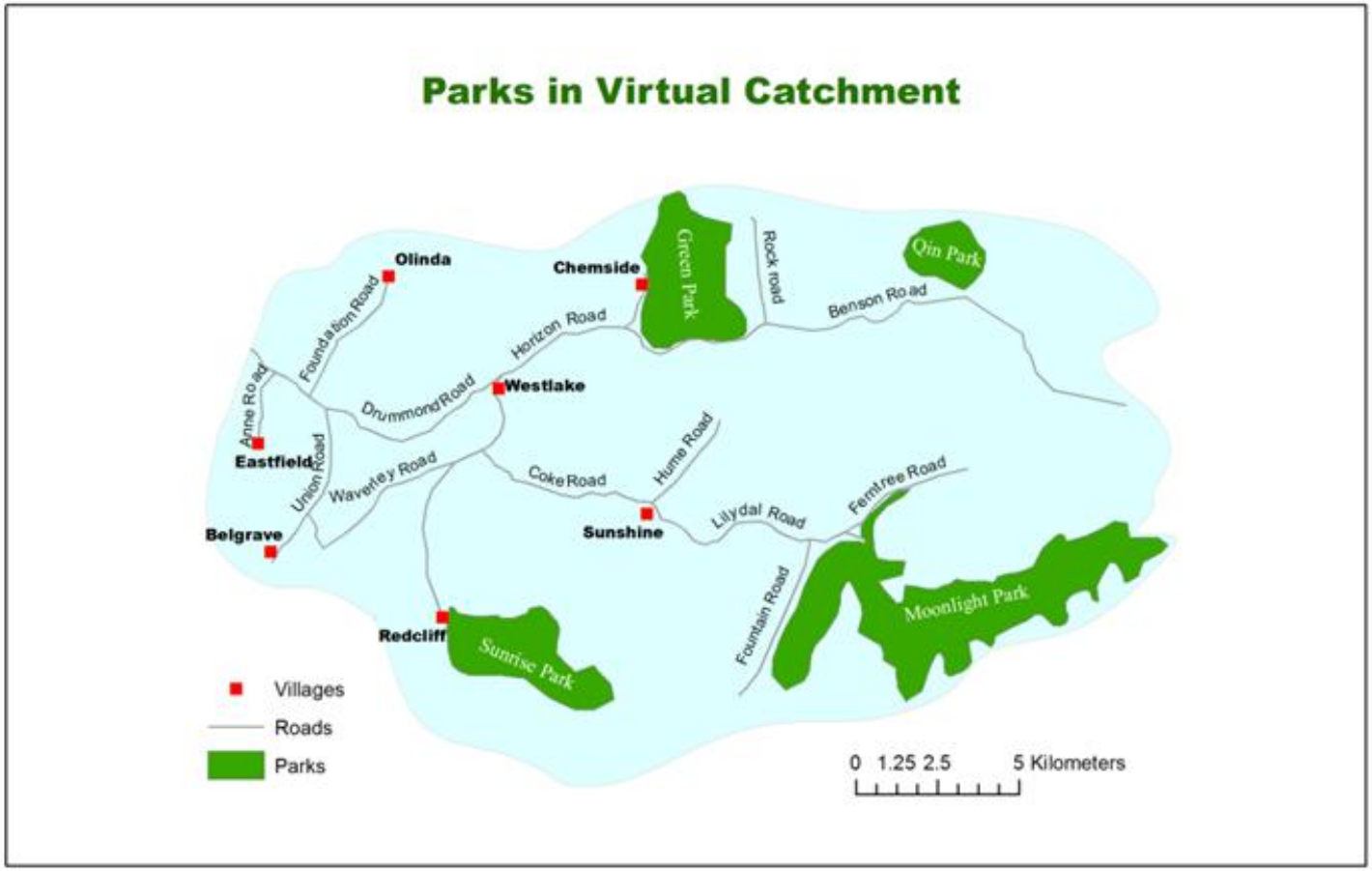
- Easily recognisable.
- Show hierarchy among geographical features.
- Help distinguish between classes and subclasses.
- Clearly identify the feature to which it refers without ambiguity.
- Does not obscure the adjacent important features.
- Only used where necessary.
- The density of annotation reflects the relative density of the distribution of the features.

Preferred positions for labelling a point feature

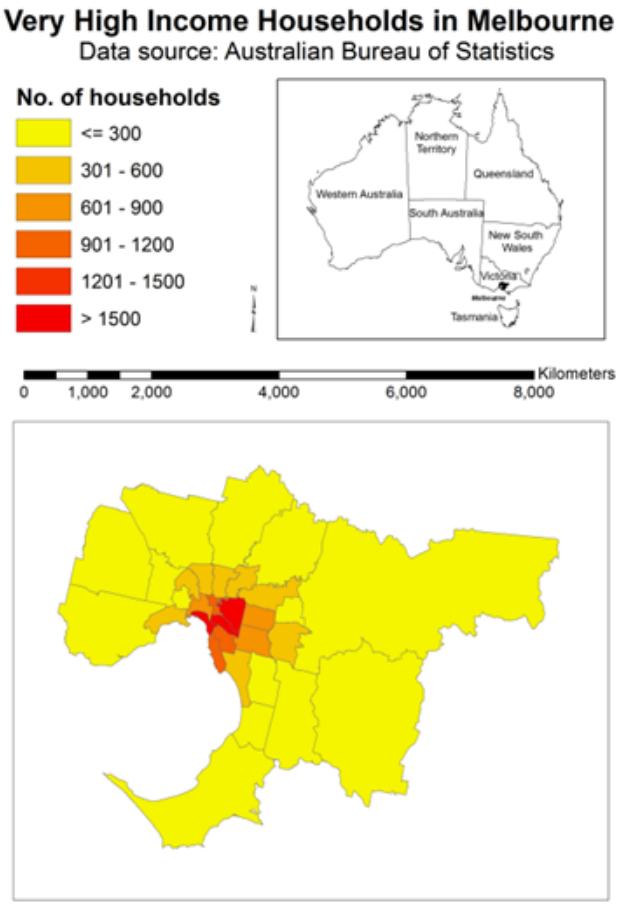


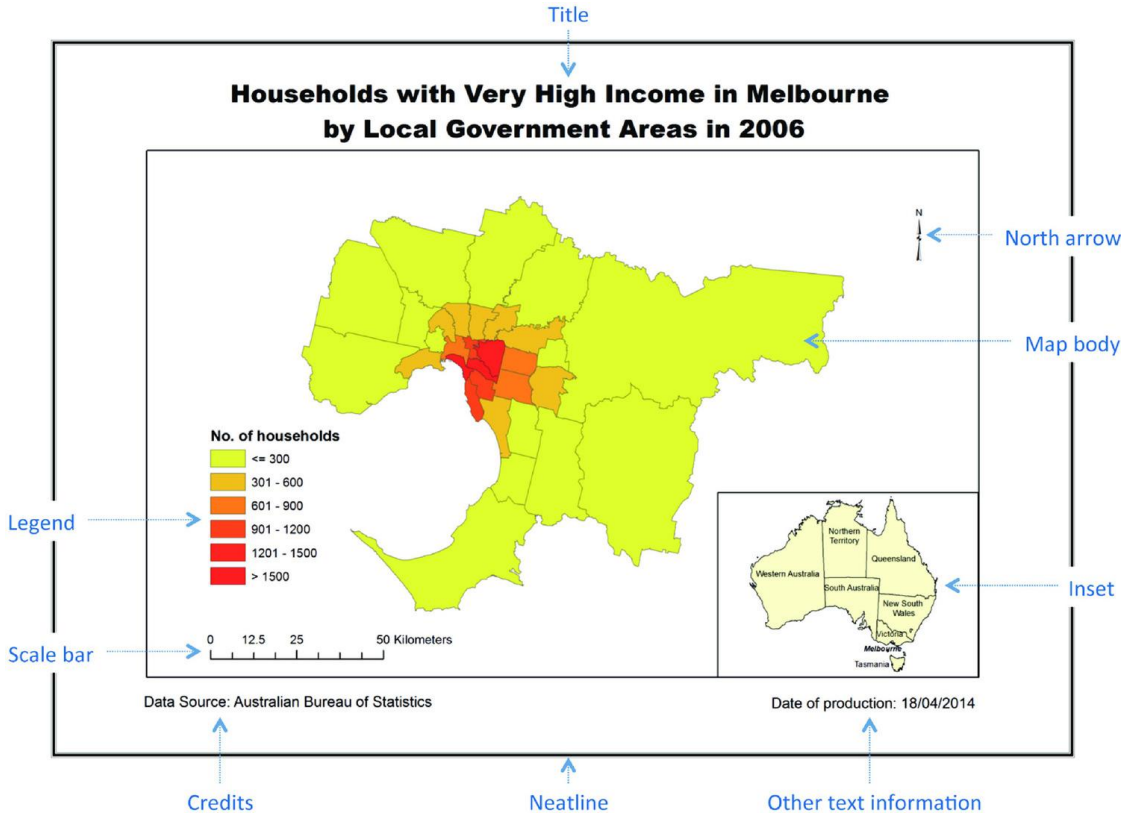
The the possible positions of annotation for a point feature ranked in order of decreasing preference.

Map lettering applied with placement rules

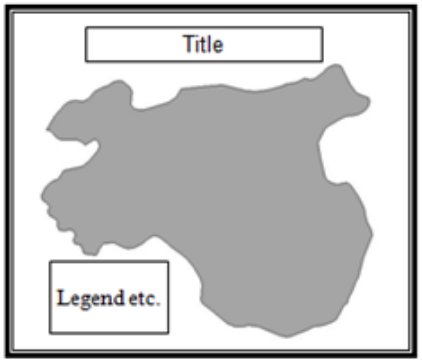


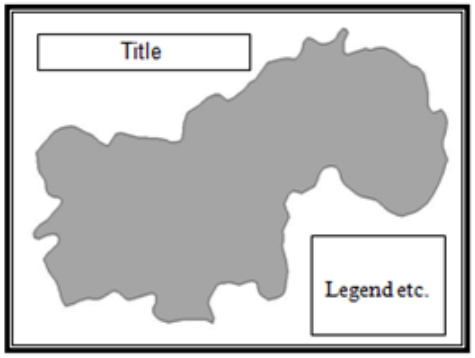
Map layout


Very High Income Households in Melbourne
Data source: Australian Bureau of Statistics
No. of households
<= 300
301 - 600
601 - 900
901 - 1200
1201 - 1500
> 1500
Northern Territory
Queensland
Western Australia
South Australia
New South Wales
Victoria
Tasmania
Kilometers
0
1,000
2,000
4,000
6,000
8,000

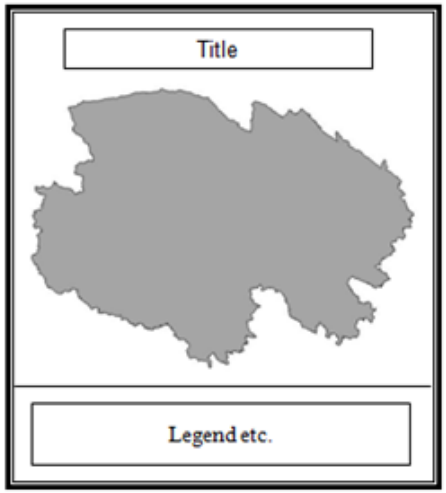

Title
Households with Very High Income in Melbourne
by Local Government Areas in 2006
North arrow
Map body
Legend
Inset
Scale bar
No. of households
<= 300
301 - 600
601 - 900
901 - 1200
1201 - 1500
> 1500
0
12.5
25
50 Kilometers
Data Source: Australian Bureau of Statistics
Date of production: 18/04/2014
Credits
Neatline
Other text information

Effective map layout design

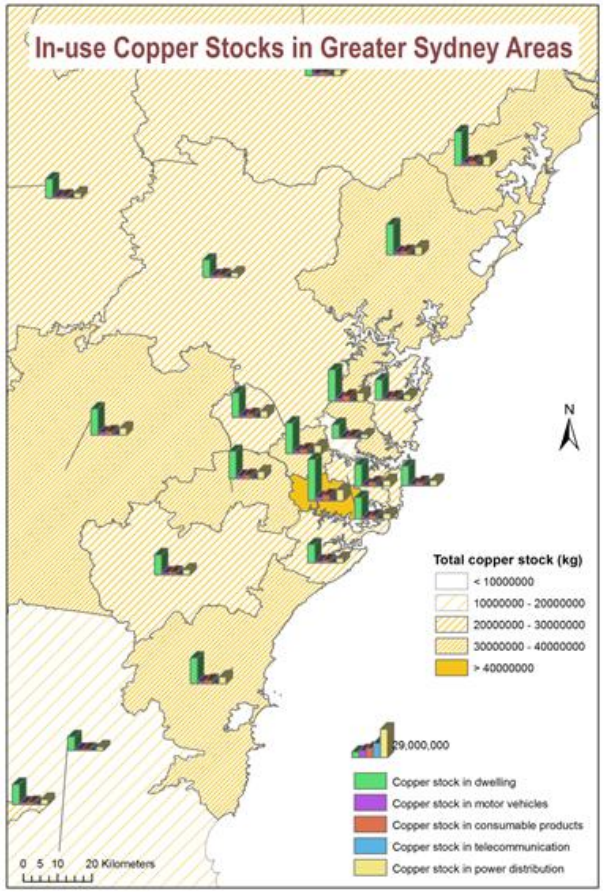






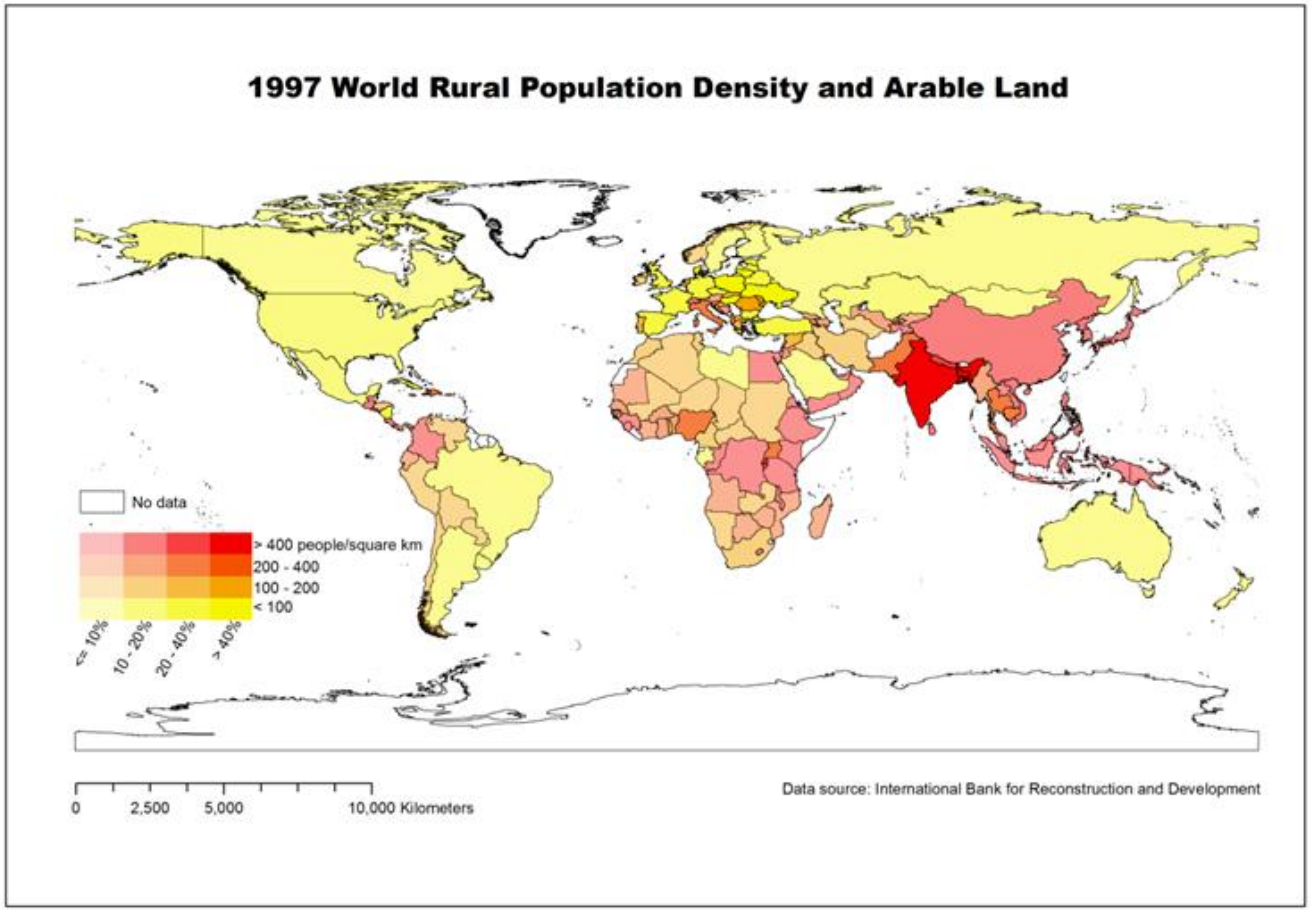


Multivariate mapping

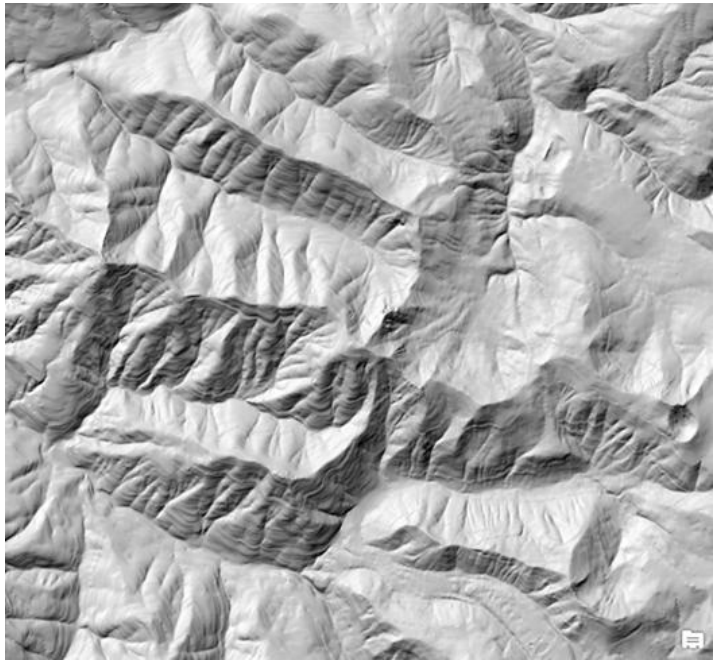


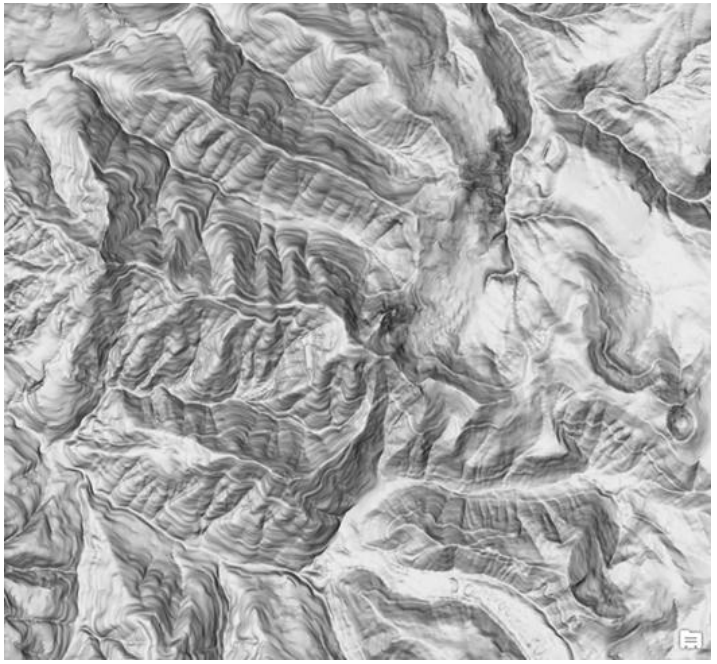
Multivariate mapping involves showing two or more attributes or variables of geographical features simultaneously.

Bivariate choropleth map



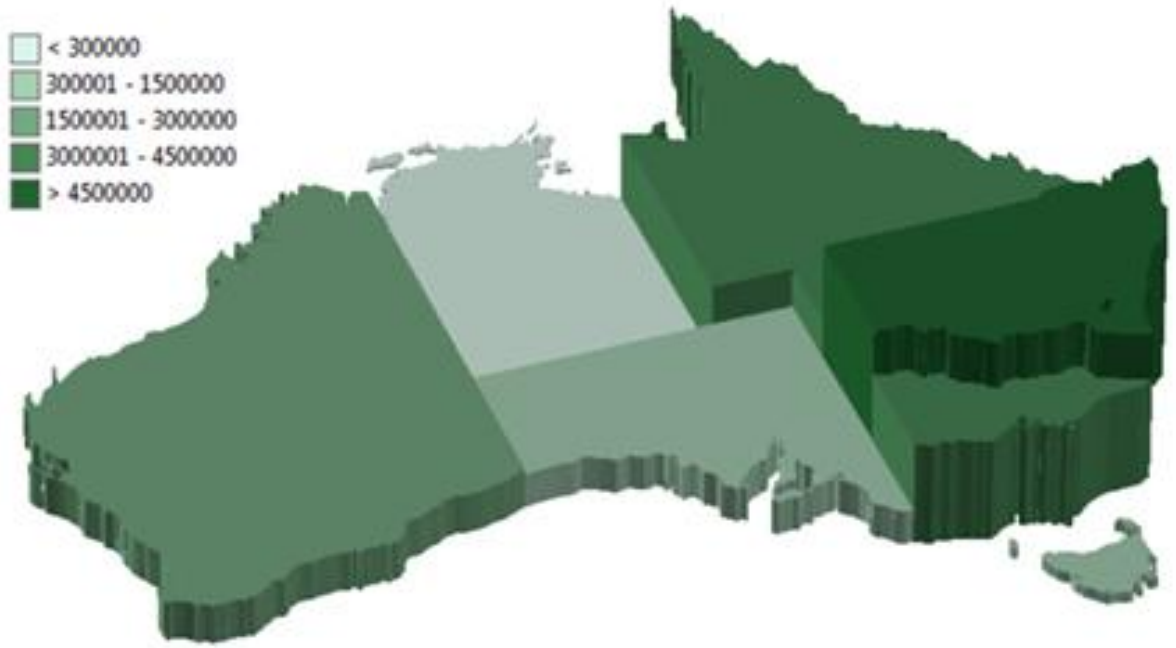
3D perspective view



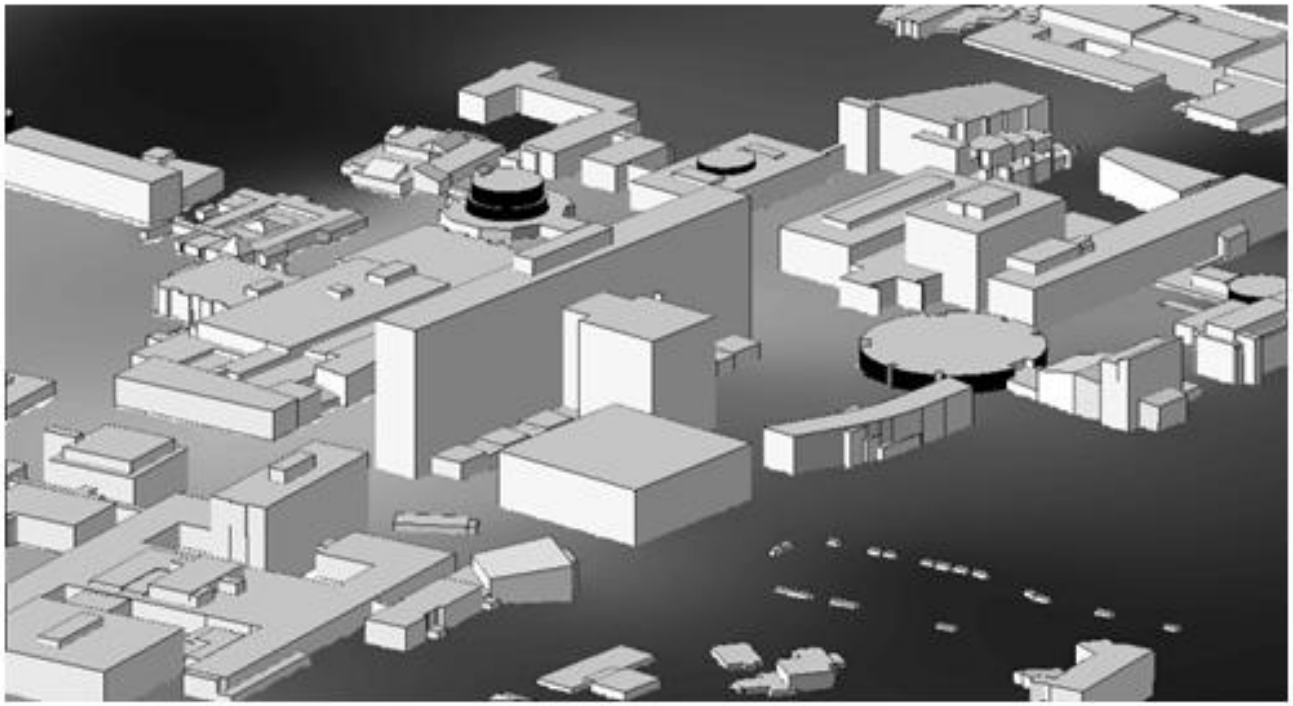


A 2.5D representation can be considered as a two-dimensional phenomenon with the third dimension representing a single attribute value e.g. elevation above sea level, air temperature, precipitation and population density.

Stepped statistical surface



3D virtual model



The annual mean $PM_{2.5}$ distribution is mapped in a 3D virtual model of Clayton Campus, Monash University. Darker colours on the ground indicate higher levels of $PM_{2.5}$

Animated mapping

- Animated mapping uses animation to show changes over **time, space** or **attributes**.
- Involves processing or playing back a sequence of individual maps, referred to as a track.
- A map layer in a track is called a frame.
- Keyframing technique is often used.
- The most characteristic frames that form an outline of the most important movements are created first.
- The computer then interpolates the ‘in-between’ frames.
- A distinction is made between temporal and non-temporal cartographic animations.

Temporal animation

- Show changes in the locational or attribute components of spatial data through maps as a time-lapse.
- Spatial data must have a time component.

Lava flow and earthquake time series (3 hr window) during May 2018 in the island of Hawaii.
https://geoxc-apps2.bd.esri.com/Analysis/HawaiiLavaFlow/

Global map of earthquakes in 2019 with an exaggerated depth.
https://geoxc-apps4.bd.esri.com/earthquakes-viz/index.html

Non-temporal animation

- Show changes caused by other variables other than time

  e.g. changes along a path, through the landscape, and with varying data presentation methods or view properties.

- Fly-through animation - provide the appearance of flying through a 3D landscape by showing a series of oblique views of the landscape.

Animation of the March 22, 2014 mudslide near Oso, Washington
https://mediaspace.esri.com/media/t/1_3xtzh7nw

Lidar masspoint data used to create a fly through of a road tunnel
https://geoxc-apps4.bd.esri.com/schiedam-tunnel/index.html

Lidar derived landscape tour of the Jack and Laura Dangermond Preserve, California, USA
https://geoxc-apps.bd.esri.com/DangermondPreserve/VirtualTour/index.html

Homework and Resources

- Read Chapter 5 of course textbook: Spatial Visualisation.
- Take the quiz to test your basic understanding (on Canvas).

Further resources

Esri ArcGIS tutorial: Introduction to Cartography (useful resource, more in-depth, worth looking at few topics for interest)

https://introduction-to-cartography-learngis.hub.arcgis.com/